DecisionRL: Enhancing Decision Calibration through Reinforcement Learning

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Research Question

As large language models evolve from question answerers to decision makers, efficient and reliable decision-making becomes increasingly important. While System 2-style models emphasize deliberate, multi-step reasoning, System 1-style models aim to provide fast, intuitive judgments. Jev introduces a complementary paradigm of structured probabilistic decisions, producing explicit probability distributions over candidate actions instead of lengthy reasoning trajectories. Our project investigates the role of **Reinforcement Learning for Calibrated Decisions (RLCD)** in improving decision accuracy and probability calibration beyond supervised fine-tuning, with a focus on opportunities to improve its optimization.

Related Work

Jev introduces System 1-style probabilistic decision-making and RLCD, while **Microsoft-Decision-1** demonstrates the potential of efficient decision models. Within the open-source ecosystem, **Kev** and **Laya** illustrate different training approaches for typed-decision models: Kev primarily combines supervised learning with post-hoc calibration, whereas Laya incorporates RLCD to improve probability calibration. These approaches motivate our investigation of what RLCD adds to supervised training and how its optimization can be improved. Our project will develop a reproducible two-stage training framework to investigate the benefits of RLCD beyond supervised fine-tuning and evaluate strategies for improving decision accuracy and probability calibration.

Planned Approach

We plan to develop **DecisionRL**, a decision model based on **EuroBERT**, through supervised fine-tuning (SFT) followed by RLCD. We will build our own training and evaluation pipeline to compare an RLCD baseline with proposed optimization variants. The work will comprise two components:

- **Data Construction and Curation.** We will collect and curate decision datasets with hard and soft labels, using a common processing pipeline for training and evaluation. Hard-labeled data will support supervised fine-tuning to establish foundational classification capabilities, while soft-labeled data will provide probabilistic supervision for RLCD. We will maintain separate training and test splits to support consistent comparisons across training stages and optimization variants.
- **Two-Stage Training and RLCD Optimization.** Starting from EuroBERT, we will perform supervised fine-tuning on hard-labeled decision data to obtain our own SFT checkpoint. We will then establish an RLCD baseline based on Laya's RLCD formulation, using soft-label supervision. Using the same SFT checkpoint, we will compare this baseline with improved RLCD optimization strategies for more accurate and better-calibrated decisions.

Evaluation Plan

Evaluation will compare our SFT checkpoint, the RLCD baseline, and our improved RLCD method on the same held-out hard-label benchmarks (e.g. Banking77) and soft-label benchmarks (e.g. Typed Decisions). **Accuracy**, **Macro F1 Score**, **Expected Calibration Error (ECE)**, and **Brier Score** will measure decision quality and probability calibration. Comparing SFT with the RLCD baseline will quantify RLCD's incremental gains; comparing the baseline with our improved method, initialized from the same SFT checkpoint and using the same training data and comparable training budgets, will test the effectiveness of the proposed optimization.

Required Resources and Division of Responsibilities

Resources: Two NVIDIA H100 GPUs and 10 GB of storage are planned for training and evaluation.

Preliminary Division of Responsibilities: Both team members will jointly participate in project discussions, research planning, and RLCD methodological design. Yanyun LIU will primarily contribute to data collection, development of the data cleaning pipeline, and quality control, while Taihua Chen will focus on implementing the two-stage training pipeline and optimizing the model and learning algorithm.